Build guide 4: axes, electronics and spindle

DIY CNC epoxy-granite mill · October 2026

6. Axes

Y (table).

1. Bolt the Y rails (full 600 mm, centerd 20 mm behind the base split) to their risers (M4 x 16, about 4 N·m). Tighten the first rail against a straightedge, then set the second parallel with a dial indicator on a block of the first (within 0.01-0.02 mm over the travel).
2. Bolt the BK12 (fixed bearing, rear) and BF12 (front) to their M6 inserts (M6 x 40, about 6-8 N·m) with the screw installed. Align the screw parallel to the rails in both directions with a dial indicator on the nut. Shim under a support if needed; the screw should spin freely by hand along the whole length.
3. Mount the DSG16H nut housing under the table (shim to height), bolt the table to the four blocks and the housing, and check for binding at both ends of travel. Fit the motor bracket, the 8 x 10 coupler and the closed-loop motor.

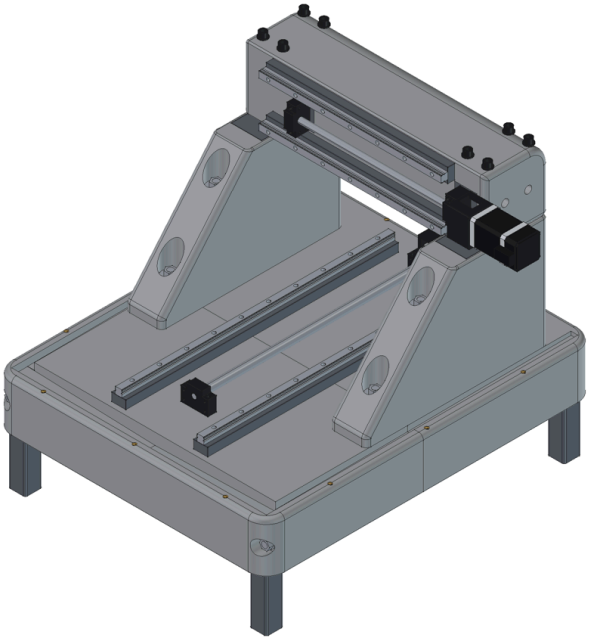
X (beam).

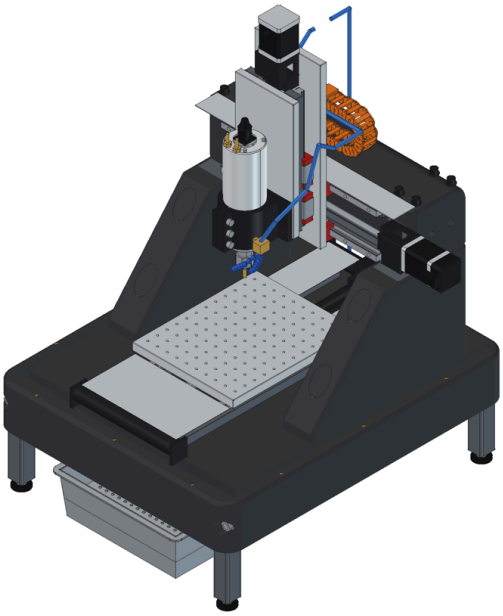
1. Rails onto the X risers as for Y (100 mm spacing).
2. BK10 and BF10 onto the beam pads: M6 x 35 with **two DIN 433 small washers under each head** (they fit the support's counterbore), so the screws stop short of the tube behind the pad. Align and check as for Y.
3. Bolt the X carriage plate to the four blocks and the nut housing; motor bracket, 8 x 8 coupler, closed-loop motor.

Z.

1. Z risers onto the Z plate: tap the risers M5 and screw them on from the back of the plate (no inserts needed in aluminum). Rails on the risers, and the four Z blocks on the front of the X carriage.
2. BK10 / BF10 and the Z screw on the carriage, nut housing on the Z plate, open-loop motor on top.
3. Spindle clamp on the Z plate, spindle in the clamp. Tram it later (section 9).

Lubricate every block with the grease gun before moving anything far.





7. Electronics and spindle

- **Controller:** FluidNC board with external drivers. CL57T drivers for X and Y (closed loop), DM542T for Z, 48 V supply trimmed to about 46 V (the CL57T maxes out at 48 V), 24 V logic.

- **Steps:** with drivers at 1600 pulses/rev: Y (SFU1605, 5 mm lead) 320 steps/mm; X and Z (SFU1204, 4 mm lead) 400 steps/mm.
- **Spindle:** VFD for 120 V now (limits it to about 1.5 kW), the kit's 220 V VFD later on 240 V. Use a dedicated 20 A circuit and shielded spindle cable, earthed at the VFD end. Set the VFD for the spindle's nameplate: 400 Hz, 220 V, 24,000 rpm.
- **Spindle cooling:** water pump in a 10-20 L bucket of distilled water with 10-20 % RV antifreeze, switched by a relay so it runs whenever the spindle runs. Never run the spindle dry.
- **Safety:** wire the E-stop to cut driver enable and the VFD run signal in hardware. Fit the homing switches, then the tool setter.